Total No. of Questions : 8]

SEAT No. :

PC-1797

[Total No. of Pages :2

[6353]-116

T.E. (Electronics & Computer Engineering)

SOFTWARE MODELING AND DESIGN

(2019 Pattern) (Semester - II) (310355 A) (Elective - II)

Time : 2½ Hours] ***[Max. Marks : 70***

Instructions to the candidates:

1) ***Answer Q.1 or Q.2, Q.3 or Q.4, Q5 or Q.6, Q7 or Q8.***
2) ***Figures to the right indicate full marks.***
3) ***Assume suitable data, if necessary.***

Q1) a) What is activity diagram? Write benefits of activity diagrams. **[6]**

b) What are some limitations or challenges associated with using activity diagrams? **[6]**

c) What are the basic components of a sequence diagram? Explain in details. **[8]**

OR

Q2) a) Describe the difference between association roles and link names in a collaboration diagram. **[6]**

b) What is a state diagram, and what is its purpose in software engineering? **[6]**

c) Explain the key differences between state diagrams, sequence diagrams, and activity diagrams in software modeling. **[8]**

Q3) a) Explain Object Oriented Constraints Language. **[8]**

b) Explain Method Design Using UML Activity Diagram. **[8]**

OR

Q4) a) Describe Object Relation Mapping, Table Class Mapping. **[8]**

b) Describe Macro-Level Design Process. **[8]**

P.T.O.

Q5) a) Explain General Responsibility Assignment Software Patterns. **[8]**

b) Explain Types of design patterns in brief. **[8]**

OR

Q6) a) Describe the Facade pattern and its use in providing a simplified interface to a complex system. **[8]**

b) Explain the difference between structural design patterns and other categories of design patterns, such as creational and behavioral patterns. **[8]**

Q7) a) Describe the role of software architecture in shaping the structure, behavior, and performance of a software system. **[9]**

b) Designing Client/Server Software Architecture. **[9]**

OR

Q8) a) Explain Designing Component-Based Software Architectures. **[9]**

b) Explain Designing Concurrent and RealTime Software Architectures. **[9]**

Total No. of Questions : 8]

SEAT No. :

PB4401

[Total No. of Pages : 2

[6262]-117

T.E. (Electronics and Computer Engineering)

SOFTWARE MODELING AND DESIGN

(2019 Pattern) (Semester - II) (Elective - II) (310355 A)

Time : 2½ Hours] ***[Max. Marks : 70***

Instructions to the candidates:

1) ***Answer Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8.***
2) ***Figures to the right indicate full marks.***
3) ***Assume suitable data if necessary.***

Q1) a) What is an activity diagram? Explain the basic elements of an activity diagram. **[6]**

b) What are some limitations or challenges associated with using activity diagrams? **[6]**

c) Explain the basic components of a sequence diagram. What are the Differences between an Activity diagram and a Flowchart? **[8]**

OR

Q2) a) Describe the difference between association roles and link names in a collaboration diagram. **[6]**

b) What is a state diagram, and what is its purpose in software engineering? Explain Triggers and Ports related to state diagram. **[6]**

c) Explain the key differences between state diagrams, sequence diagrams, and activity diagrams in software modeling. **[8]**

Q3) a) What is object-oriented constraints language? Explain Object Oriented Constraints Language. **[8]**

b) Explain Method Design Using UML Activity Diagram. **[8]**

OR

Q4) a) Describe Object Relation Mapping and Table Class Mapping. **[8]**

b) Describe Macro-Level Design Process. **[8]**

Q5) a) Explain General Responsibility Assignment Software Patterns. **[8]**

b) Explain Types of design patterns in brief. **[8]**

OR

Q6) a) Describe the Facade pattern and its used in providing a simplified interface to a complex system. **[8]**

b) Explain the difference between structural design patterns and other categories of design patterns, such as creational and behavioral patterns. **[8]**

Q7) a) Describe the role of software architecture in shaping the structure, behavior and performance of a software system. **[9]**

b) Explain designing Client/Server Software Architecture. **[9]**

OR

Q8) a) Explain designing Component-Based Software Architectures. **[9]**

b) Explain designing Concurrent and Real Time Software Architectures. **[9]**

◀▶ ◀▶ ◀▶